BA 336

Essentials of Information Technology

Fall 2025

Instructor:	Shaw Millerman, MBA
Email:	m.a.s@pdx.edu
Section:	BA 336 (Fall '26), Section: **HB1**
Class Meetings:	In-person**:** Wednesdays from 9:00am - 10:50pm, KMC 560
Online Office Hours:	Available after class or by appointment.
Attendance	Attendance is required. You must also log into CANVAS multiple times during the week to keep on track with the weekly readings and assignments.
Requirements	Textbooks and high speed internet. PSU computer labs are equipped for this course and available for free student use. If you lack access to any required tech, contact the PSU Library about borrowing options.

Instructor and Course Information

Communication: I try to be timely with email responses. If you email during the week you will usually receive a response within 24 hours. Please use your PSU email and include BA 336 in the subject line. Do not use the CANVAS Inbox.

Course Description

This course explores the critical role of data and information in supporting business organizations. Students will grasp the essential relationships between data, business processes, and information technology systems. As a survey course, it emphasizes the ability to leverage both quantitative and qualitative data for analysis, decision-making, and communication. Key topics include data interpretation, effective thinking skills, and practical examples of real-world, workplace applications. A primary focus of this course is on learning Excel skills to analyze data in support of decision making in the workplace.

Course Objectives

This course focuses on 1) the many ways that information technology enhances business processes, and 2) how this technology, creatively applied, allows for business leaders to optimize for productivity, refine business processes, and develop new services and products. The result is a more competitive, successful business, with more satisfied customers, and happy engaged employees. After successful completion of this course, students will be able to:

1. Recognize the significance of management information systems to organizational decision-making and success.
2. Review computer hardware and types of software, including open source software and some technical considerations in buying new technology.
3. Distinguish the roles of different types of information systems (e.g., transaction processing systems, decision support systems, etc.) used in supporting business processes.

4. Explain the importance of data/information management to the business organization and database technologies used to maintain data.
5. Show proficiency with basic decision support technologies.
6. Learn, build, and demonstrate Microsoft Excel skills and concepts
7. Become familiar with WordPress as a tool for building websites
8. Become familiar with the Google suite of applications through hands-on activities

Required Texts



Information Systems: A Manager's Guide to Harnessing Technology (Version 9.1.1), Gallaugher, Flat World Knowledge, 2023, available at this link: https://students.flatworldknowledge.com/engage/courses/2615282 or from the PSU bookstore. **Important Note: You only need the Online Access Only version to be successful in our class.**

Beginning Excel 2019, Authors: Brown, Lave Puncochar, Romey, Schatz, Schneider, and Shingledecker; Open Oregon Pressbooks Publisher. Link to the free OER text: https://openoregon.pressbooks.pub/beginningexcel19/

Suggested Weekly Flow

Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
Review Canvas Content, Read Textbook Chapters	Read Textbook Chapters Watch Videos	Read Textbook Chapters Watch Videos, Work on Practice Activities	Work on Projects and Assignments	Work on Projects and Assignments	Work on Projects and Assignments	Projects, and Assignments due by 11:59 pm

Participation and Engagement

This is a working class, your presence, preparation, and contribution matter. Plan to attend every meeting, arrive on time, and engage in discussions and in-class activities. You have two no-questions-asked absences as grace, beyond that, unexcused absences limit make-ups for in-class work. If you must miss class, notify me in advance when possible and complete the posted make-up (if applicable) within 7 days. One missed quiz is already covered by the automatic drop. More than 2 absences will impact your participation grade.

Course Assignments and Projects

Excel Assignments (42 pts) and Excel Projects (12 pts)

Excel is one of the most widely used tools in business and one of the most valuable to master. In this course, you'll build a strong foundation in Excel over the first six weeks, progressing from beginner to intermediate-level skills through a series of hands-on assignments (worth 42 points total). You'll also complete two Excel Projects. These projects are your chance to apply

what you've learned so far to practical scenarios, just like you would on the job. Each project is worth 6 points, for a total of 12 points.

Comparative Analysis of Critical Terms (8 pts)

This assignment helps you build fluency in key industry terms while practicing clear, concise writing. You'll choose several terms from the textbook and compare their definitions to those generated by ChatGPT in two different roles. The goal is to reflect on how language, context, and purpose shape the outputs of generative AI.

Midterm (6 pts)

The midterm exam will take place during Week 5 and will assess your understanding of the material covered in the first half of the course. This includes content from the textbook, in-class lectures, discussions, and Excel assignments. This will be an in-class exam consisting of 20 multiple choice, fill in the blank, and true/false questions, as well as 3 short answer questions.

Information Systems in the News (4 pts)

This discussion-based assignment connects course concepts to current events. You'll choose a recent news article related to technology or information systems and explain how it relates to topics we've covered.

Google Sheets vs. Excel (4 pts)

This assignment revisits a prior Excel task, but this time in Google Sheets. It's designed to help you understand the functional differences between the two platforms and develop comfort working in browser-based spreadsheet tools

Case Study (8 pts)

In this case-based group assignment, you'll take on the role of a junior analyst tasked with uncovering insights from one year of sales data at a fictional office and tech supply company. Working in Google Sheets or Excel, you'll clean and organize raw data, create pivot tables and charts, and use your findings to make strategic business recommendations. This project emphasizes analytical thinking, collaboration, and using data to support real-world decision-making.

Prompt Engineering Assignment (8 pts)

This assignment builds on your introduction to generative AI by challenging you to refine and improve AI prompts in a business context. You'll take a vague prompt and improve it step by step, learning how clarity, context, and structure impact the quality of AI responses. The assignment is tied to a realistic workplace scenario and encourages iteration, reflection, and critical thinking.

Final Exam (8 pts)

The Final Exam is 20 multiple choice and 3 short answer questions. The Final Exam covers material from chapters 15, 16, 17, and 18, in the Information Systems textbook. The Exam is open book, open slides, and open notes. The final exam will take place Wednesday, December 10, 9:00am - 10:50am.

Course Evaluation & Grading

Scored Items	Point Value Each	Total Point Value
Excel Activities • **6 Guided** • **6 Practice** • **6 Challenge**	1 point each 2 points each 4 points each	42 points
Comparative Analysis of Critical Terms • **Chap 1, 6, 10** • **Chap 16, 17, 18**	4 points each	8 points

Excel Projects	6 points each	12 points
Midterm	6 points	6 points
Google Sheets vs. Excel	4 points	4 points
Information System in the News	4 points	4 points
Prompt Engineering	8 points each	8 points
Case Study	8 points	8 points
Final Exam	8 points	8 points
Total		**100 Points**

How Feedback Works

- All scores live in Canvas Gradebook, comments appear on the assignment page and in SpeedGrader.
- **Instructor-graded items:** Brief rubric comments tied to criteria, plus quick notes on common errors and next steps.
- **When:** Auto-graded items, immediate. Instructor-graded items, within 7 days of the due date, sooner when possible.
- **Revisions:** Only when explicitly allowed on the assignment page, policies vary by assignment. Read comments before asking for a regrade, then follow the **Grade Review Requests** policy if you still see an issue.

Course Schedule

Week/Topic	Course Content and Activities

Week 1 Intro to Business and Information Systems Excel Basics	• Required reading: o Syllabus o Information Systems Chapter 1: Setting the Stage: Technology and the Modern Enterprise o The Importance of Data Literacy and Data Storytelling • Required videos: o An Introduction to BA336 and Information Systems (video) o Art Schneider's Excel Assignments Tutorial o How to Deal with Big Tech o Judge Rules Google Has An Illegal Monopoly • Apply: o Excel Chapter 1 (*Due Sunday by 11:59 pm*) ▪ CH1 Merchandise City Sales Data (1.1-1.4) ▪ PR1 Medical Office Budget (1.5) ▪ SC1 Sales and Inventory (1.6)
Week 2 Introduction to Hardware Excel Computations	• Required reading: o Computer Hardware and Components (PDF) o The Rise and Fall of Intel o Information Systems Chapter 6: Moore's Law and More • Required videos: o Computer Hardware (video) o Relative Reference Formulas o Payment (PMT) Function Tutorial • Apply: o Excel Chapter 2 (*due Sunday by 11:59 pm*) ▪ CH2 Personal Budget (2.1-2.4) ▪ PR2 Lawn Care (2.5) ▪ SC2 Hotel (2.6) o (Extra Credit) LinkedIn Profile Project (*due Sunday by 11:59 pm*)
Week 3 Networks, the Internet, and Websites Excel Formulas and Functions	• Required reading: o Information Systems Chapter 10: Platforms, Network Effects, and Competing in a Winner-Take-All World o Comparative Analysis of Critical Terms: Assignment Guide o Critical Terms Wordbank (Information Systems Chapters 1, 6, 10, 15) o Excel Chapter 3: Formulas, Functions, Logical and Lookup Functions • Required videos: o E-book videos (embedded in Information Systems textbook) • **Apply**: o Comparative Analysis of Critical Terms #1 (Information Systems Chapters 1,6,10, 15) (*due Sunday by 11:59 pm*) o Excel Chapter 3 (*due Sunday by 11:59 pm*) ▪ CH3 Gradebook and Parks (3.1-3.4) ▪ PR3 Redding (3.5) ▪ SC3 MidasCoffee (3.6)

Week 4 Information Systems and Social Media Excel Charts	• Required reading: o Information Systems Chapter 11: Social Media, Peer Production, and Leveraging the Crowd • Required videos: o The Evolution of Social Media from 2004 - 2024 o Open AI May be Developing Its Own Social Media Platform • **Apply**: o Excel Chapter 4 (*due Sunday by 11:59 pm*) ▪ CH4 Charting (4.1-4.4) ▪ PR4 Historical Data (4.5) ▪ SC4 Sales (4.6) o Begin Excel Project 1 - Concepts from Excel Chapters 1-3 (*due week 5 but start this week*)
Week 5 Introduction to Software Excel Tables and Pivot Tables	• Required reading: o Computer Software (PDF) o Information Systems Chapter 15: Understanding Software: A Primer for Managers • Required videos: o Computer Software (video) o E-book videos (embedded in textbook) • **Apply**: o Excel Chapter 5 (*due Sunday by 11:59 pm*) ▪ CH5 HR Report (5.1-5.4) ▪ PR5 Canyon Trails (5.5) ▪ SC5 Dynamite Customer Sales (5.6) o Excel Project 1 (*due Sunday by 11:59 pm*) o Midterm (textbook chapters 1, 6, 10, and 11)
Week 6 Software and Cloud Services Excel 3D Formulas and Multiple Sheets	• Required reading: o Networks and the Internet (PDF) o Information Systems Chapter 16: Software in Flux: Open Source, Cloud, Virtualized, and App-Driven Shifts o Comparative Analysis of Critical Terms Assignment Guide o Critical Terms Wordbank (Information Systems Chapters 16, 17, 18, Emerging Tech) • Required videos: o Networks and the Internet (video) o E-book videos (as listed in weekly CANVAS content) o Introduction to Prompt Engineering • **Apply**: o Comparative Analysis of Critical Terms #2 (Information Systems Chapters 16, 17, 18, Emerging Tech) (*Due Sunday by 11:59 pm*) o Excel Chapter 6 (*due Sunday by 11:59 pm*) ▪ CH6 Personal Budget (6.1-6.4) ▪ PR6 Pacific NW Sports Team (6.5) ▪ SC6 National Parks (6.6)

Week 7 Data Management and Google Drive	• Required reading: o Introduction to Relational Database (PDF) o Information Systems Chapter 17: Data and Competitive Advantage: Databases, Analytics… o Google Sheets vs. Microsoft Excel (Zapier Article) • Required videos: o Database Tutorial for Beginners o Flat File vs Database o Relational Database Overview • **Apply**: o Google Sheets vs. Microsoft Excel (Repeat Chapter Practice 2.5 using Google Sheets) o Information Systems in the News
Week 8 AI and Emerging Technologies Part 1	• Required reading: o Information Systems Chapter 18: Artificial Intelligence o 3 Quantum Computing Myths Debunked o Quantum Infographic • Required videos: o Quantum Computing is the Next Revolution o Ransomware Attack Explained o The Difference Between Viruses, Worms & Trojans (video) o Viruses, Worms, and Botnet Explained (video) o E-book videos (as listed in CANVAS) • **Apply**: o Excel Project 2 (*Due Sunday by 11:59 pm*)
Week 9 Prompt Engineering and Emerging Technologies Part 2	• Required reading: o Prompt Engineering Is Not the Future (An AI Rewrite) o What is Blockchain? • Recommended reading: o Case Study Project Instructions (due week 10) • Required videos: o What is Blockchain? The Best Explanation of Blockchain Technology o Is Prompt Engineering a Joke or the Future of AI Craftsmanship? • **Apply**: o Applied Prompt Engineering Project (*Due Sunday by 11:59 pm*)

Week 10 Case Study and Wrap-up	• Required reading: Review for Final Exam • Required videos: The Great Hack (in-class) • **Apply**: o Case Study Project (*Due Sunday by 11:59 pm*) o **Final Deadline for all approved late assignments is Monday by 11:59 pm**
Week 11 Final Exam	• **Apply**: o **Final Exam** (textbook chapters 15, 16, 17, and 18). The Final Exam will be updated when the exam schedule is released. o Open book, open notes, open slides

Grading Scale

Letter Grade	Percent Conversion
A	100 - 93
A-	92.999 - 90
B+	89.999 - 87
B	86.999 - 83
B-	82.999 - 80
C+	79.999 - 77

Letter Grade	Percent Conversion
C	76.999 - 73
C-	72.999 - 70
D+	69.999 - 67
D	66.999 - 60
F	59.999 - 0

General Requirements for Submitted Work:

- Points may be deducted for poor presentation, accuracy, or incompleteness.
- **Excel work must be submitted as .xlsx files.**
- ALL assignment submissions or emails should follow this example:
    - BA336U
    - Student's name
    - Examples
        - **Assignment Example**: *BA 336 - Shaw Millerman - Excel Project.xlsx*
        - ***Email subject line**: BA 336 - Excel Project Question*

Course Policies and Student Support

AI Use Policy

AI tools (such as ChatGPT) may be used in this course only when explicitly allowed on an assignment. When permitted, students must cite AI assistance just as they would any other source. Using AI to complete work without disclosure, or in ways that misrepresent your own effort, is considered academic dishonesty under PSU's Student Conduct Code. If you are unsure whether AI use is acceptable, ask before you act.

Schedule Changes

The schedule and assignments in this syllabus are subject to change if needed to support learning or respond to unexpected events. Any changes will be announced in advance through Canvas.

Late Work

Due dates are important in this class as we move quickly through the material. Also, the professional skill of meeting deadlines is valuable. Due dates are firm but exceptions include a Five-Day Late "Window:" Assignments may be submitted up to five days late with a 12% point deduction. That means the highest grade you can get is a B+. After that, submissions are closed. Late Work Note: For emergencies beyond the two-week window, contact your instructor ASAP. Extensions may be granted on a case-by-case basis. Technical issues on any machine will not be accepted as an excuse for incomplete work. Start assignments with plenty of time to finish and/or time to address technical issues.

Communication Channels

Canvas is the official channel for course announcements, deadlines, and grades. Check it daily. Email is the best way to contact me directly about personal or course-related questions. For quick updates or clarifications, always look to Canvas first.

Respect and Diversity Statement

This classroom is a community built on respect. We bring different backgrounds, perspectives, and experiences, and that diversity strengthens our learning. I ask that you engage with curiosity, professionalism, and civility. Disagreement is welcome when expressed respectfully. Our shared goal is to create an environment where every student feels included, valued, and able to contribute fully.

University Policies

Disability Statement:

PSU values diversity and inclusion; we are committed to fostering mutual respect and full participation for all students. My goal is to create a learning environment that is equitable, useable, inclusive, and welcoming. If any aspects of instruction or course design result in barriers to your inclusion or learning, please notify me. The Disability Resource Center (DRC) provides reasonable accommodations for students who encounter barriers in the learning environment. If you have, or think you may have, a disability that may affect your work in this class and feel you need accommodations, contact the Disability Resource Center to schedule an appointment and initiate a conversation about reasonable accommodations. The DRC is located in 116 Smith Memorial Student Union, 503-725-4150, drc@pdx.edu, https://www.pdx.edu/drc. If you already have accommodations, please contact me to make sure that I have received a faculty notification letter and discuss your accommodations.

Title IX Reporting Policy

As your instructor, I'm committed to fostering a safe, respectful, and professional learning environment. Discrimination or harassment of any kind, including sexual harassment, has no place in our class or on campus. You may report concerns to the **Office of Equity and Compliance or the Office of the Dean of Student Life**.

Please note that, as a faculty member, I am required to report any instances of sexual harassment, sexual violence, or other prohibited discrimination. If you prefer to speak with someone confidentially, you can reach out to a designated confidential employee. To learn more about your rights and resources, complete the **Understanding Sexual Misconduct and Resources** module in Canvas.

Sexual Harassment

Sexual harassment is prohibited by University policy, Oregon State System of Higher Education Administrative Rules, and by State and Federal law. See the following document for more details: Prohibited discrimination and sexual harassment policy